

2110203 COM ENG MATH II

SIGNAL PROCESSING

PART

1

OBJECTIVES & STUDENT OUTCOMES

2110203 Com Eng Math II (Signal part):

Objectives: To be able to understand and apply fundamental signal processing including Fourier transform and analysis

CPSO: (a.2) an ability to apply knowledge of computer mathematics to solve computer engineering problems

CEN Lab : Utilize Math Tools

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LESSON PLAN

WEEK	MONDAY 13:30–15:30	TOPIC	THU 13:30-15:30	TOPIC
1	10/8/2024	INTRO, SIGNALS, TRANSFORM	10/10/2024	INTRO, SIGNALS, TRANSFORM
2	10/15/2024	CONVOLUTION	10/17/2024	CONVOLUTION
3	10/22/2024	CTFT		
4	10/29/2024	CTFT		
5	11/5/2024	מכונות	11/7/2024	חשבון SIGNAL
6	11/12/2024	DTFT	11/14/2024	DTFT
7	11/19/2024	DFT		
8	11/26/2024	APPLICATIONS		
9	12/3/2024	מכונות		
10	12/11/2024	חשבון SIGNAL		

Signal Part:

-Mid-part exam 11/7/2024 (1.5-2 hours)

Topics : Intro, Signals, Transformations, Convolution and Fourier Series

-Final-part exam 12/11/2024 (2 hours)

Topics : Fourier Transform, Discrete-time Fourier Series/Transform, DFT and Applications

BOOKS

- Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab., **Signals and Systems**, 2nd Edition, Prentice Hall (1996)
- Samir S. Soliman and Mandyam D. Srinath., **Continuous and Discrete Signals and Systems**, 2nd Edition, Prentice Hall (1998)

1. Signals & systems / Alan V. Oppenheim, Alan S. Willsky, with Hamad Nawab.

By: Oppenheim, Alan V., Upper Saddle River, N.J. : Prentice Hall, c1997, xxx, 957 p. : ill., 25 cm.
Language: English, Database: CU Library

Subjects: System analysis; Signal theory (Telecommunication)

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Location	Call No.	Status
Engineering Library	QA402 O63 1997 c.3	CHECK SHELVES

[Show More \(2\)](#)

1. Continuous and discrete signals and systems / Samir S. Soliman, Mandyam D. Srinath.

By: Soliman, Samir S., Upper Saddle River, NJ : Prentice Hall, 1998 xvii, 525 pages : illustrations ; 25 cm.
Language: English, Database: CU Library

Subjects: Signal theory (Telecommunication); Signal Processing -- Mathematics; System analysis

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Location	Call No.	Status
Engineering Library	TK5102.9 S65 1998	CHECK SHELVES

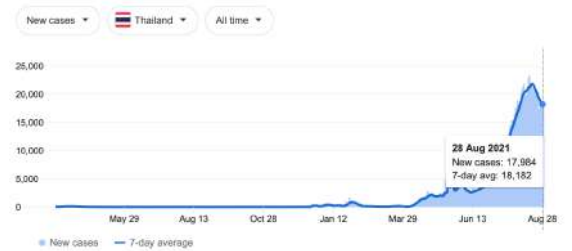
SIGNALS EXAMPLES



Statistics

New cases and deaths

From JHU CSSE COVID-19 Data · Last updated: 2 days ago



Reference : <https://youtu.be/x4nO5ygarM0>

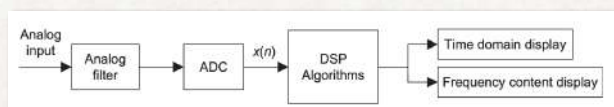
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BASIC DIGITAL SIGNAL PROCESSING EXAMPLES

- Digital Filtering → Clean "Noise"



- Signal Frequency (Spectrum) Analysis



Li Tan and Jean Jiang (2019), *Digital Signal Processing Fundamentals and Applications* (3rd Edition)

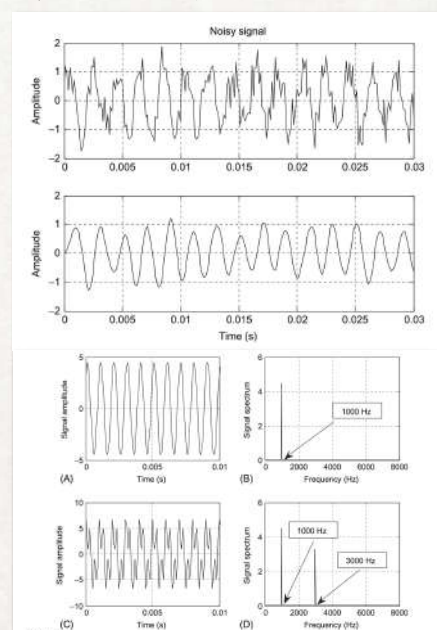


FIG. 1.5 Audio signals and their spectra. (A) 1000-Hz audio signal. (B) 1000-Hz audio signal spectrum. (C) Audio signal combining 1000 and 3000-Hz frequency components. (D) Audio signal spectrum containing 1000 and 3000-Hz frequency components.

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SIGNALS FREQUENCY DOMAIN

- Noise removal



- <https://youtu.be/keYzcEHugx8>

- Image enhancement →



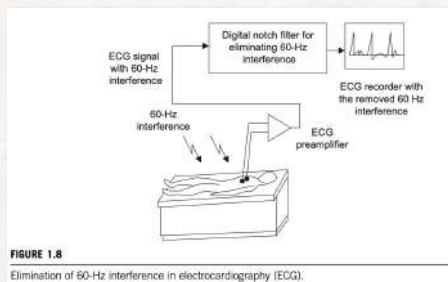
- <https://www.osapublishing.org/oe/fulltext.cfm?uri=oe-22-1-618&id=276513>

Thin cloud removal from single satellite images by Jun Liu, et al. (2014)

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DIGITAL SIGNAL PROCESSING IN REAL WORLD APPLICATIONS

- Digital Crossover Audio System
- Interference cancellation in electrocardiography



- Speech coding and compression

Li Tan and Jean Jiang (2019), *Digital Signal Processing Fundamentals and Applications (3rd Edition)*

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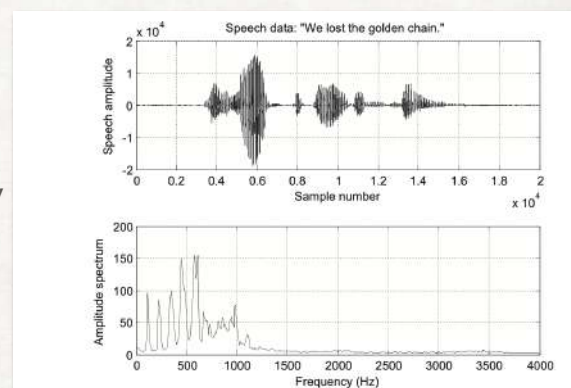
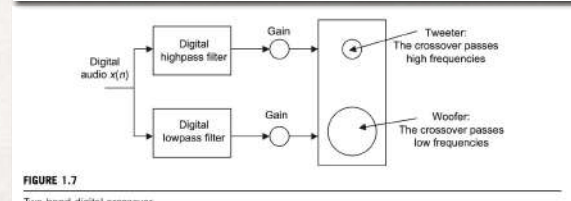


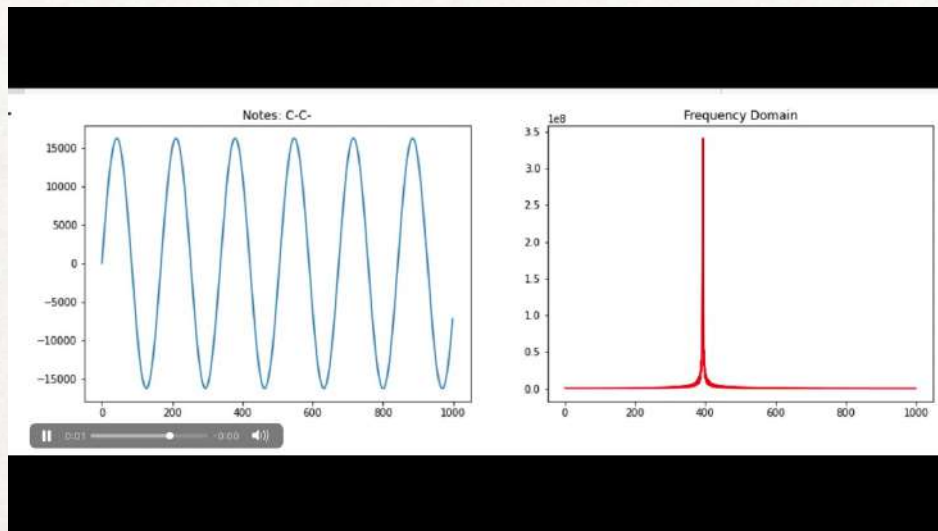
FIGURE 1.6

Speech samples and speech spectrum.



DEMO

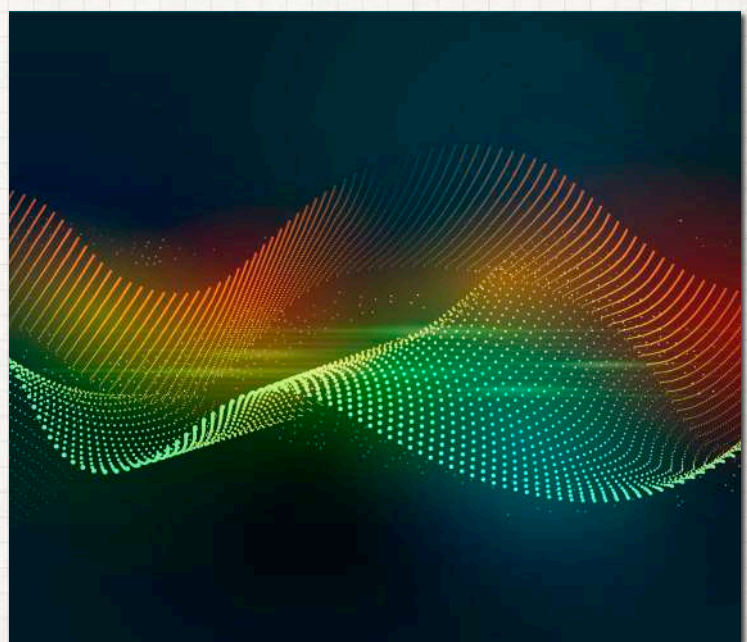
FREQUENCY DOMAIN



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LECTURE 01

REPRESENTING SIGNALS



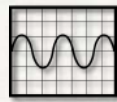
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INTRODUCTION

Signal represented by $x(t)$ / $f(x,y)$
 ↑
 independent time

Examples of signals?

- Human voice
- Television pictures
- Atmosphere temperature
- What we see.. EM energy.. Light!



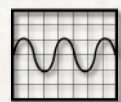
Temperature
windspeed
Speech
Light intensity

Physical quantities

"Measurable"

TRANSDUCER

Time-varying current / voltage signals



Function representation of one or more independent variables (x-y coordinates, time space domain)

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CONTINUOUS-TIME VS. DISCRETE-TIME SIGNALS

Continuous signal:

- (1) Continuous-time signal - independent variable is continuous

- Example: telephone / audio signals or atmosphere pressure

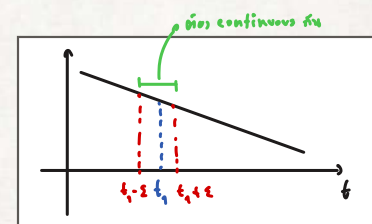
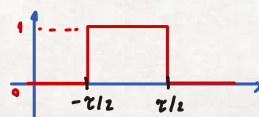
- (2) Signal $x(t)$ is continuous if $x(t)$ is continuous of all t

- At instant t_1 and small positive ϵ , $t_1 + \epsilon$ and $t_1 - \epsilon$ denoted as t_1^+ and t_1^- , respectively. If $x(t_1^+) = x(t_1^-) = x(t_1)$, we said $x(t)$ is continuous at $t = t_1$

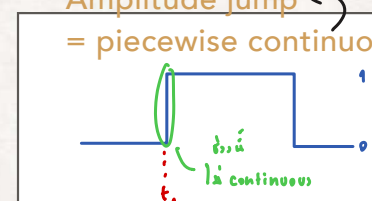
- Otherwise, it is discontinuous at t_1 and amplitude has jump at that point.

Continuous-time signal is not continuous:

$$1 \text{ rect}\left(\frac{t}{\tau}\right) = \begin{cases} 1, & |t| < \frac{\tau}{2} \\ 0, & |t| > \frac{\tau}{2} \end{cases}$$



Amplitude jump
= piecewise continuous



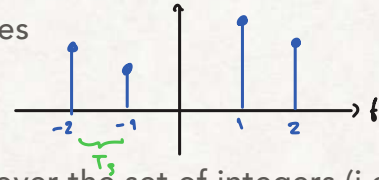
A rectangle pulse signal

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CONTINUOUS-TIME VS. DISCRETE-TIME SIGNALS

- If the independent variable takes on only discrete values

$$t = kT_s$$



- Where T_s is a fixed positive real number and k ranges over the set of integers (i.e., $k = 0, \pm 1, \pm 2$, etc.)
- The corresponding signal $x(kT_s)$ is called a discrete-time signal : $x[k] / x[n]$ ^{integer}
- Example - the amount of a loan payment in k th month, number of children born on a specific day - denoted as $x[k]$

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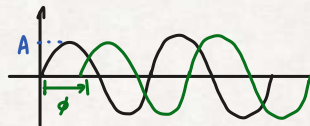
CONTINUOUS-TIME SIGNALS

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PERIODIC VS. APERIODIC

Հոսանքի ժամանակ :  (Time-Domain)

- Any continuous-time signal satisfies condition:
 - $x(t) = x(t + nT)$ $n = 1, 2, 3, \dots$ $x(t)$ is periodic
 - where $T > 0$ is a constant known as the fundamental period
- An example of a periodic signal: sinusoidal function
 - $x(t) = A \sin(\omega_0 t + \phi)$ A.K.A. Sine wave
 - Where A is an amplitude, ω_0 is a radian frequency (rad/s) and ϕ is an initial phase angle (in radian)
 - Fundamental period $T = \frac{2\pi}{|\omega_0|}$



$\omega_0 : 2\pi f = \frac{2\pi}{T}$
 A.K.A. ω_0

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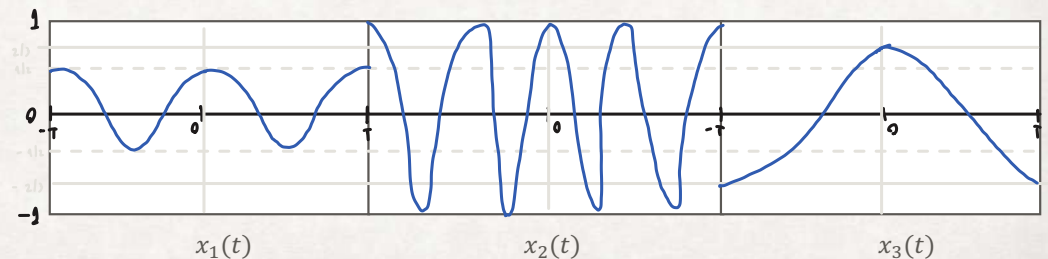
PERIODIC VS. APERIODIC

If $x(t)$ is periodic with fundamental period T , then $x(t)$ is also periodic with period $2T, 3T, \dots$

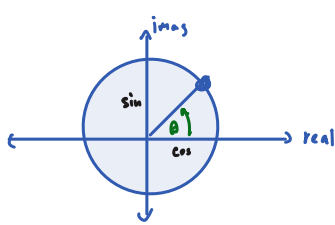
ω_0 : radian frequency $\omega_k = k\omega_0$ kth harmonic

Waveforms at each harmonic are distinct!

- $x_1(t) = \frac{1}{2} \cos 2\pi t$
- $x_2(t) = \cos 4\pi t$
- $x_3(t) = \frac{2}{3} \cos \pi t$



Հոսանքի ժամանակ 2T (-T to T)



$e^{j\theta} = \cos(\theta) + j\sin(\theta) \Rightarrow \text{Complex Exponential}$
 $\begin{matrix} \cos(\omega t) & + & j\sin(\omega t) \\ \text{real part} & & \text{imag part} \end{matrix}$
 $\begin{bmatrix} k \uparrow \text{freq} \\ k \downarrow \text{am} \end{bmatrix}$

PERIODIC VS. APERIODIC

EXERCISE

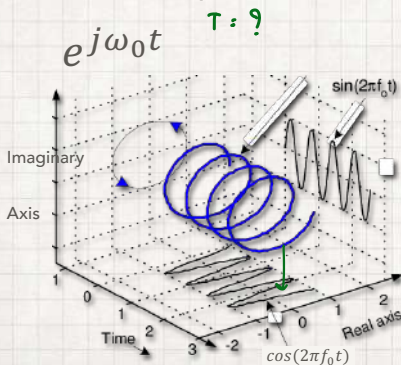


SINE WAVE VS. COMPLEX EXPO FUNCTION
 $e^{jx} = \cos x + j\sin x$

Example #1:

$\omega_k = k\omega_0$ (base freq. $\rightarrow$ harmonic freq.)
 $T = ?$

Harmonically related continuous-time exponentials are sets of complex exponentials with fundamental frequencies that are all multiples of a single positive frequency ω_0 . Find the fundamental period of this signal.

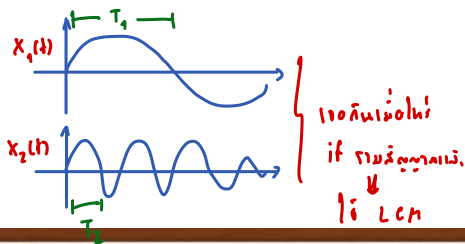


$\phi_k(t) = e^{jk\omega_0 t}$ where $k = 0, \pm 1, \pm 2, \dots$
 $\text{Euler's: } e^{j\phi} = \cos \phi + j\sin \phi$

$x(t) = e^{j(k\omega_0 t)} = \cos(k\omega_0 t) + j\sin(k\omega_0 t)$
 $\therefore \text{Fundamental Period (T)} = \frac{2\pi}{|\omega_k|} = \frac{2\pi}{|k\omega_0|}$

Richard G. Lyons, Understanding Digital Signal Processing's Frequency Domain (2001)

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PERIODIC VS. APERIODIC

Is the sum of two periodic signals periodic?

- The sum of two or more periodic signals may $\neq$ periodic.
- Consider $x(t)$ and $y(t)$ with fundamental period T_1 and T_2 , respectively. Considering the sum

$z(t) = ax(t) + by(t)$ is periodic?
 $x(t) = x(t + kT_1)$, $y(t) = y(t + lT_2)$

$x(t) = x(t + kT_1)$, $y(t) = y(t + lT_2)$

$\text{So } z(t) = ax(t + kT_1) + by(t + lT_2) \Rightarrow \text{sum is periodic!}$

In order z to be periodic with period T , one needs

$z(t) = ax(t + T) + by(t + T) = ax(t + kT_1) + by(t + lT_2)$

We must have $T = kT_1 = lT_2$ where k and l are integers.

Note: T must be the LCM of T_1 and T_2 .

Note: for Trigon

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
 $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
 $\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$

$1) x_1(t) = \sin \frac{2\pi}{3} t \sim T_1 = \frac{2\pi}{|\omega_1|} = \frac{2\pi}{(\frac{2\pi}{3})} = 3 \text{ sec}$
 $2) x_2(t) = \sin \frac{2\pi}{5} t \cdot \cos \frac{4\pi}{3} t = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$
 $= \frac{1}{2} [\sin(\frac{2\pi}{5} + \frac{4\pi}{3})t + \sin(\frac{2\pi}{5} - \frac{4\pi}{3})t] + \frac{1}{2} \sin(4-8)$

$\therefore T = kT_1 = lT_2 \Rightarrow k \cdot \frac{2\pi}{3} = l \cdot \frac{2\pi}{5}$
 $k \cdot 5 = l \cdot 3 \Rightarrow k = 3, l = 5$
 $\therefore T = 3 \cdot 3 = 9 \text{ sec}$

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$$p(t) = v(t) \cdot i(t)$$

$$\frac{v^2(t)}{R} = \underline{i^2(t) \cdot R}$$

↳ Magnitude for signal "2"

- $$E_{2L} = \int_{-L}^L |x(t)|^2 dt$$

Total energy $t \in (-\infty, \infty)$: $E = \lim_{L \rightarrow \infty} \int_{-L}^L |x(t)|^2 dt$

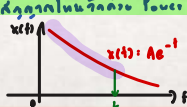
- Average power (P) $\rightarrow \frac{\text{Signal Energy}}{\text{Time Interval}}$

$$P = \lim_{L \rightarrow \infty} \left(\frac{1}{2L} \int_{-L}^L |x(t)|^2 dt \right)$$

- $0 < E < \infty \rightarrow x(t)$ is an energy signal with zero power
- $0 < P < \infty \rightarrow x(t)$ is a power signal with infinite energy

Note အမှတ်လက်မှတ်ပေး Power / စွမ်းအင် Energy ?

Energy Signal



$$\left[\text{Total Energy} = \lim_{L \rightarrow \infty} \int_{-L}^L |x_1(t)|^2 dt \right]$$

$$= \lim_{L \rightarrow \infty} \int_{-L}^L A^2 e^{-2t} dt$$

Power Signal


$$\begin{aligned} \text{Total Energy: } & \lim_{L \rightarrow \infty} \int_{-L}^{+L} |\chi_2(t)|^2 dt & \frac{d}{dt}(-g(t)) = -2 \\ & & \frac{d}{dt}(-g(t)) = -2 \cdot \frac{d}{dt}t \\ & : \lim_{L \rightarrow \infty} \left(\int_{-L}^{+L} A^2 dt + \int_{-L}^{+L} A^2 e^{-2t} dt \right) & \frac{d}{dt}(-g(t)) = -2 \cdot \frac{d}{dt}t \\ & & \frac{d}{dt}(-g(t)) = -2 \cdot \frac{d}{dt}t \end{aligned}$$

Note in Power

$$P = \frac{\text{Total Energy}}{\text{Time Interval}} = \lim_{L \rightarrow \infty} \left(\frac{1}{2L} \int_{-L}^{+L} |x_1(t)|^2 dt \right)$$

$$= \lim_{L \rightarrow \infty} \left(\frac{1}{2L} \cdot \frac{A^2}{2} \right) = 0 \quad (\text{as } L \rightarrow \infty)$$

$\therefore$ an Energy $\uparrow$, Power $\uparrow$ $\Rightarrow$ "Energy Signal"

$$L: \lim_{L \rightarrow \infty} A^2 \int_{t=0}^{t=L} e^{-2t} \frac{A(-2t)}{-2} dt$$

$$L: \lim_{L \rightarrow \infty} -\frac{1}{2} A^3 \int_0^L d(-2t)$$

$$L: \lim_{L \rightarrow \infty} -\frac{1}{2} A^3 \left[e^{-2t} + c \right]_{t=0}^{t=L}$$

$$L: \lim_{L \rightarrow \infty} -\frac{1}{2} A^3 \left[e^{-2(L)} - e^{-2(0)} \right]$$

$$L: \lim_{L \rightarrow \infty} -\frac{1}{2} A^3 \left(e^{-2L} - 1 \right)$$

$\frac{d}{dt} = \frac{d(-2t)}{-2}$
 (substitute d)
 (substitute $t=0$ and integrate to L)
 via $\int e^x dx = e^x +$

សិទ្ធិរក្សា Energy ថា

$$\begin{aligned}
 p &= \frac{E_{\text{energie}}}{T_{\text{zeit}}} = \lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t (N_2(t))^\alpha dt \\
 &= \lim_{t \rightarrow \infty} \left(\frac{1}{t} \left(A^2 t - \frac{1}{2} A^2 t^2 e^{-2t} + \frac{1}{2} A^2 \right) \right) \\
 &= \lim_{t \rightarrow \infty} \left(\frac{1}{2} A^2 - \underbrace{\frac{1}{4} \frac{A^2}{L} e^{-2t}}_{\substack{\text{für } t \rightarrow \infty \\ \rightarrow 0}} + \underbrace{\frac{1}{4} \frac{A^2}{t}}_{\substack{\text{für } t \rightarrow \infty \\ \rightarrow 0}} \right) \\
 &= \frac{1}{2} A^2 \rightarrow \text{un Power in, Energie fließt}
 \end{aligned}$$

$$\begin{aligned} &: \lim_{L \rightarrow \infty} \left(A^2 \int_{-L}^{-1} dI + -\frac{1}{2} A^2 \int_{-L}^{-1} e^{-2I} d(-2I) \right) \\ &: \lim_{L \rightarrow \infty} \left(A^2 (0 - (-L)) + -\frac{1}{2} A^2 (e^{-2(-L)} - e^{-2(0)}) \right) \quad \left(e^{-2L} : e^{-\infty} = 0 \right) \\ &: \lim_{L \rightarrow \infty} \left(A^2 L + -\frac{1}{2} A^2 (e^{-2L} - 1) \right) \\ &: \lim_{L \rightarrow \infty} \left(A^2 L - \frac{1}{2} A^2 e^{-2L} + \frac{1}{2} A^2 \right) : \infty \end{aligned}$$

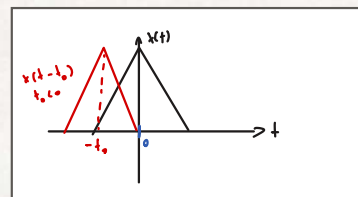
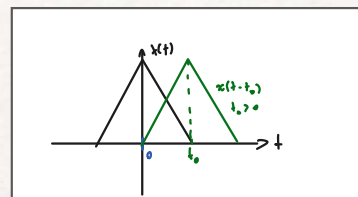
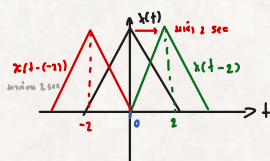
$\lim_{L \rightarrow \infty} A^2 L = \infty$
 $\lim_{L \rightarrow \infty} -\frac{1}{2} A^2 e^{-2L} = 0$
 $\lim_{L \rightarrow \infty} \frac{1}{2} A^2 = \frac{1}{2} A^2$

$\therefore E \rightarrow \infty$ (unbound)

Power Signal

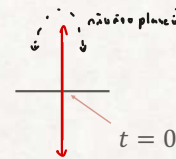
SHIFTING OPERATION

- Signal $x(t - t_0)$ is time shift version of $x(t)$
 - If $t_0 > 0$: signal is delayed by t_0 seconds
 - If $t_0 < 0$: signal is advanced replica of $x(t)$
 - e.g., radar, sonar, communication systems, seismic

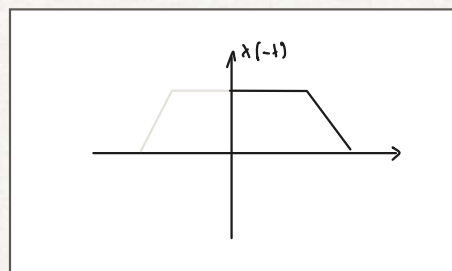
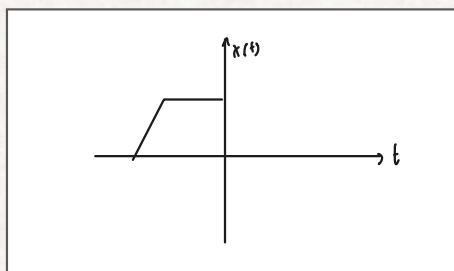


TRANSFORMATIONS OF INDEPENDENT VARIABLE

REFLECTION OPERATION



- Signal $x(-t)$: reflection of $x(t)$ about $t = 0$
- e.g., if $x(t)$: signal out of an audio recorder, then $x(-t)$: signal out of an audio player played backward.

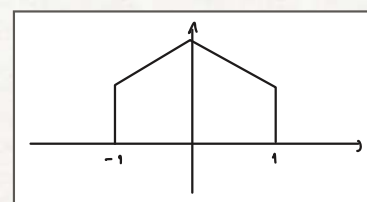


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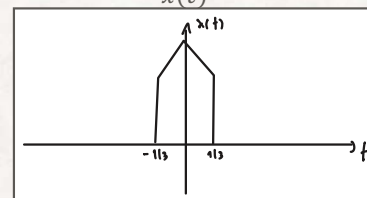
TRANSFORMATIONS OF INDEPENDENT VARIABLE

TIME-SCALING

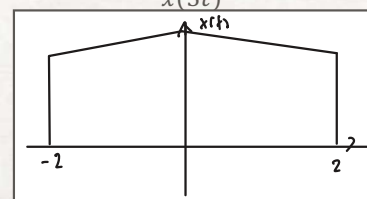
- Signal $x(t)$ is scaled by parameter η
- $x(\eta t)$: a compressed version if $|\eta| > 1$ $\text{Ex } x(t) \rightarrow x(3t)$
 an expanded version if $|\eta| < 1$ $\text{Ex } x(t) \rightarrow x(\frac{1}{2}t)$
- e.g., if $x(t)$ output of video tape recorder
 $x(3t) \rightarrow$ recording is played three times speed
 $x(\frac{t}{2}) \rightarrow$ played at half speed



$x(t)$



$x(3t)$



$x(\frac{t}{2})$

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TRANSFORMATIONS OF INDEPENDENT VARIABLE

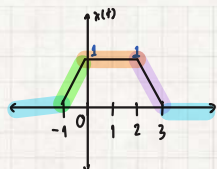
EXERCISE



Example #2:

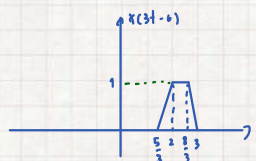
Plot $x(3t - 6)$

$$x(t) = \begin{cases} t+1, & -1 \leq t \leq 0 \\ 1, & 0 < t \leq 2 \\ -t+3, & 2 < t \leq 3 \\ 0, & \text{otherwise} \end{cases}$$



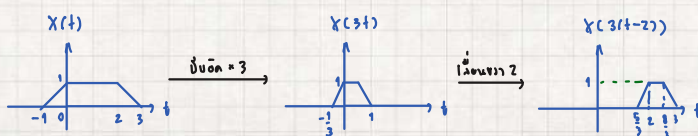
Solⁿ 1: Amplitude scaling
amplitude $u = 3t - 6$

$$\therefore X(u) = \begin{cases} u+1; & -1 \leq u \leq 0 \\ 1; & 0 < u \leq 2 \\ -u+3; & 2 < u \leq 3 \\ 0; & \text{otherwise} \end{cases} \rightarrow x(t) = \begin{cases} 3t-5; & -1 \leq 3t-6 \leq 0; \quad \frac{5}{3} \leq t \leq 2 \\ 1; & 0 < 3t-6 \leq 2; \quad 2 < t \leq \frac{8}{3} \\ -3t+9; & 2 < 3t-6 \leq 3; \quad \frac{8}{3} < t \leq 3 \\ 0; & \text{otherwise} \end{cases}$$



Note: Amplitude scaling $\therefore$ Amplitude remains the same
No amplitude scaling in $x(3t)$ but in $x(3t-6)$ (amplitude is 1)

Solⁿ 2: Time Transform and Amplitude



TRANSFORMATIONS

- For any verbal signal $x(t)$, the transformation $\alpha t + \beta$ on the independent variable can be performed as follows:

$$x(\alpha t + \beta) = x(\underbrace{\alpha}_{\text{no scale}}(t + \beta/\alpha))$$

(1) Scale by α . If α is negative, reflect about the vertical axis.

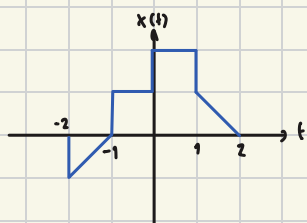
(2) shift to the right by β/α if β and α have different signs, and to the left if β and α have the same sign.

Quiz In-Class

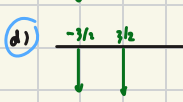
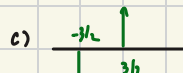
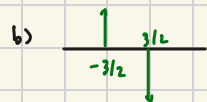
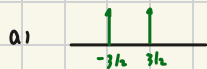
1) Find Period (T) of signal $x(t) = e^{j4\pi t}$

Solⁿ $x(t) = e^{j4\pi t} = \cos(4\pi t) + i \sin(4\pi t)$ via Euler Complex: $e^{i\theta} = \cos \theta + i \sin \theta$
 $\therefore \omega_0 = 4\pi = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{4\pi} = \frac{2\pi}{4\pi} = 0.5 \text{ sec}$

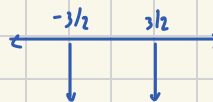
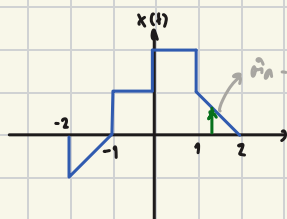
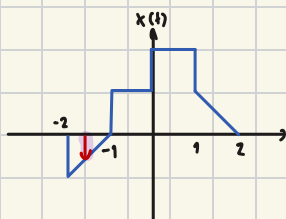
2)



Find $x(t) \left[\delta(t + \frac{3}{2}) - \delta(t - \frac{3}{2}) \right]$



Solⁿ using $x(t) = x(t) \delta(t + \frac{3}{2}) - x(t) \delta(t - \frac{3}{2})$ [via Property: $x(t) \delta(t - t_0) \Rightarrow$ value at graph]



3) Determine Fundamental Period of Signal

$z(t) = 2 \cos(10t + 1) - \sin(4t - 1)$
 $x_1(t)$ $x_2(t)$

Solⁿ using formula: $\text{LCM of } \omega_1 \text{ and } \omega_2 \text{ is } \omega_0$

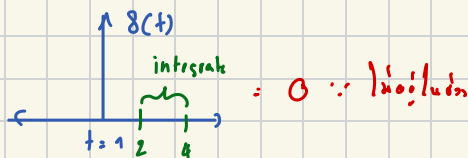
$\therefore x_1(t) = 2 \cos(10t + 1) \rightarrow \omega_1 = \frac{2\pi}{T_1} = 10 \Rightarrow T_1 = \frac{1}{5} \pi$
 $x_2(t) = -1 \sin(4t - 1) \rightarrow \omega_2 = \frac{2\pi}{T_2} = 4 \Rightarrow T_2 = \frac{1}{2} \pi$

using $\text{LCM of } z(t) \text{ is Periodic Signal} \rightarrow T_2 = k \cdot T_1 = l \cdot T_2$
 $T_2 = k \cdot \left(\frac{1}{5} \pi\right) = l \cdot \left(\frac{1}{2} \pi\right)$
 $\frac{k}{l} = \frac{5}{2} \Rightarrow \text{LCM of } 5 \text{ and } 2$

$\therefore k = 5, l = 2 \rightarrow T_2 = k T_1 = (5) \left(\frac{1}{5} \pi\right) = \pi \text{ sec}$

4) $\int_2^4 (t^2 + 2t + 1) \delta(t - 1) dt = ?$

- (a) 4
(b) 2
(c) 1
(d) 0



Note for Trigon

$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

$\cos(A \mp B) = \cos A \cos B \mp \sin A \sin B$

$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$
 $\sin A - \sin B = 2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$
 $\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$
 $\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$

OTHER PROPERTIES

- Even signal - even symmetric: $x(t)$ is identical to its time reflection about origin

$$x(t) = x(-t) \Rightarrow \text{Reflect } \text{κατάσπασμα} \text{ } \underline{\text{Εξ}} \text{ } \text{συμμετρίας cosine}$$

- Odd signal - odd symmetric

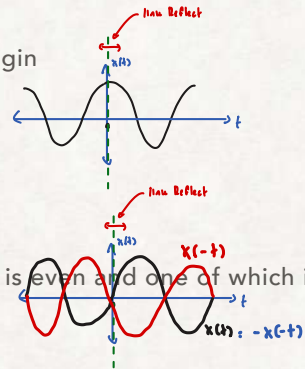
$$x(t) = -x(-t) \Rightarrow \text{Reflect } \text{κατάσπασμα} \text{ } \underline{\text{Εξ}} \text{ } \text{συμμετρίας sine}$$

Συμμετρική ή non odd-even ή

- Any signal $x(t)$ can be decomposed in to a sum of two signals, one of which is even and one of which is odd : $x(t) = x_e(t) + x_o(t)$

$$\text{Even part: } x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$\text{Odd part: } x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$



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ELEMENTARY SIGNALS

- Unit Step function (continuous-time) defined as

$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

- Ramp Function (r(t))

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Obtained by integrating unit step function

i.e., linear-sweep waveform of a cathode ray tube

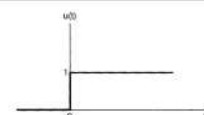
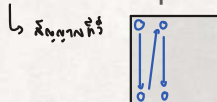


Figure 1.32 Continuous-time unit step function.

in the relationship between the discrete-time unit impulse and step functions. In particular, the continuous-time unit step is the *running integral* of the unit impulse:

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau. \quad (1.71)$$

This also suggests a relationship between $\delta(t)$ and $u(t)$ analogous to the expression for $\delta[n]$ in eq. (1.65). In particular, it follows from eq. (1.71) that the continuous-time unit impulse can be thought of as the *first derivative* of the continuous-time unit step:

$$\delta(t) = \frac{du(t)}{dt}. \quad (1.72)$$

In contrast to the discrete-time case, there is some formal difficulty with this equation as a representation of the unit impulse function, since $u(t)$ is discontinuous at $t = 0$ and consequently is formally not differentiable. We can, however, interpret eq. (1.72) by considering an approximation to the unit step $u_\Delta(t)$, as illustrated in Figure 1.33, which rises from the value 0 to the value 1 in a short time interval of length Δ . The unit step, of course, changes values instantaneously and thus can be thought of as an idealization of $u_\Delta(t)$ for Δ so short that its duration doesn't matter for any practical purpose. Formally, $u(t)$ is the limit of $u_\Delta(t)$ as $\Delta \rightarrow 0$. Let us now consider the derivative

$$\delta_\Delta(t) = \frac{du_\Delta(t)}{dt}, \quad (1.73)$$

as shown in Figure 1.34.

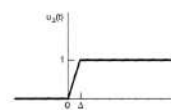


Figure 1.33 Continuous approximation to the unit step, $u_\Delta(t)$.

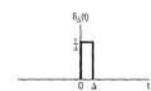


Figure 1.34 Derivative of $u_\Delta(t)$.

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ELEMENTARY SIGNALS

- Unit Step function (continuous-time) defined as

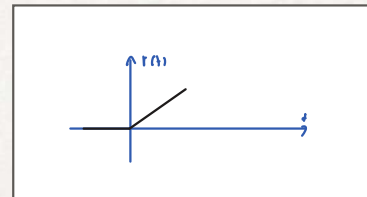
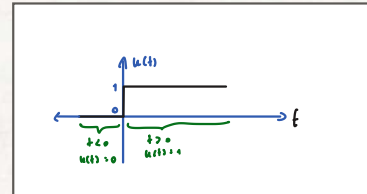
$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

- Ramp Function

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Obtained by integrating unit step function

i.e., linear-sweep waveform of a cathode ray tube



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Note: non L'Hopital $\approx$ diff un l'hopital

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

for $Sa(x = \frac{\pi}{2})$

$$x = \frac{\pi}{2} \rightarrow Sa(x = \frac{\pi}{2}) = \frac{\sin(\pi/2)}{\pi/2} = \frac{1}{\pi/2} = \frac{2}{\pi} (< 1)$$

$$\therefore \lim_{x \rightarrow 0} Sa(x) = \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = \cos(0) = 1$$

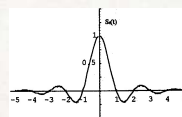
$\frac{0}{0}$ indeterminate form

ELEMENTARY SIGNALS

- Sampling function $Sa(x) = \frac{\sin x}{x}$ or $\text{sinc}(x)$

$$Sa(x) = \frac{\sin x}{x} \quad \text{and} \quad \text{sinc}(x) = \frac{\sin \pi x}{\pi x}$$

$\text{sinc}(x) = 0$ at $x = \pm \pi, \pm 2\pi, \dots$



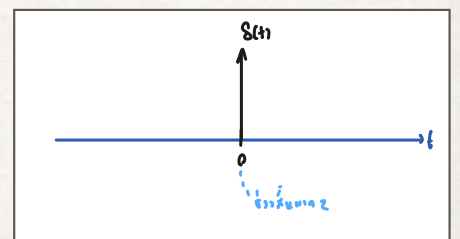
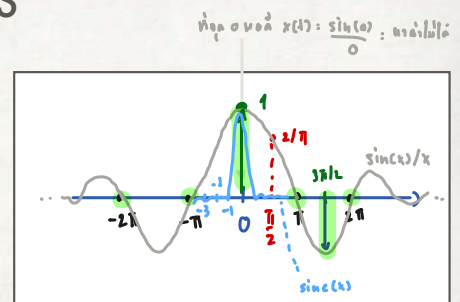
- Unit Impulse Function - or Dirac delta function or Delta function $\delta(t)$

- e.g., physical phenomena such as point sources, point charges, concentrated loads on structures, voltage or current source acting very short times

- Delta function can be defined as

$$\int_{t_1}^{t_2} x(t) \delta(t) dt = x(0) \quad : t_1 < 0 < t_2$$

- $\delta(t) \sim$ spike at origin



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ELEMENTARY SIGNALS

Note $\delta(t)$ is a representation of an impulse (an infinitesimal delay) $\approx \delta(t-t_0)$ at t_0 (an infinitesimal delay)

- Unit Impulse Function - or Dirac delta function or Delta function $\delta(t)$ has the following properties

1) $\delta(0) \rightarrow \infty$ (infinitesimal width 0)

2) $\delta(t) = 0, \quad t \neq 0$

3) $\int_{-\infty}^{\infty} \delta(t) dt = 1$ (area under the curve is 1)

4) $\delta(t)$ is an even function, i.e., $\delta(t) = \delta(-t)$ (reflect across t=0)

Note $x(t) \rightarrow x(bt)$ compressed
 $x(t) \rightarrow x(\frac{t}{b})$ expanded

3 Properties

① Sifting Property

$$\int_{t_1}^{t_2} x(t) \cdot \delta(t-t_0) dt = \begin{cases} x(t_0) & t_0 \in [t_1, t_2] \\ 0 & \text{otherwise} \end{cases}$$

Diagram illustrating the sifting property: A signal $x(t)$ is shown with an impulse $\delta(t-t_0)$ at t_0 . The integral of $x(t) \cdot \delta(t-t_0)$ over a range $[t_1, t_2]$ is zero if t_0 is outside the range, and $x(t_0)$ if t_0 is inside the range.

② Sampling Property

$$x(t) \delta(t-t_0) = x(t_0) \delta(t-t_0)$$

Diagram illustrating the sampling property: A signal $x(t)$ is shown with an impulse $\delta(t-t_0)$ at t_0 . The product $x(t) \delta(t-t_0)$ is shown as a scaled impulse $x(t_0) \delta(t-t_0)$.

③ Scaling Property

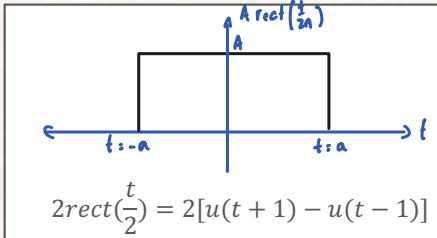
$$\delta(at+b) = \frac{1}{|a|} \delta(t+\frac{b}{a})$$

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using approximation Delta function $\rightarrow$ impulse function

OTHER TYPES OF SIGNALS

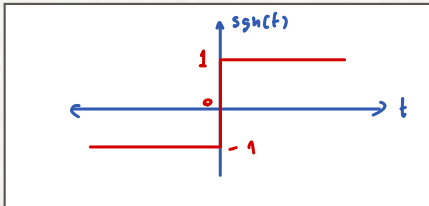
- rect : rectangular pulse signal (on-off switching)



Rectangular pulse signal extended from $-a$ to a with an amplitude A

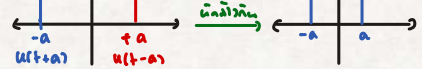
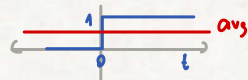
i.e., $A * \text{rect}(t/2a) = A(u(t+a) - u(t-a))$ (using Unit Step function)

- sgn (signum function) [using Unit Step and Average = 0]



$$\text{sgn}(t) = -1 + 2u(t)$$

Note Unit Step $\rightarrow$ Average > 0



Note (on Delta Function Property)

$\int_{-2}^1 (t+t^2) \delta(t-3) dt$: $\textcircled{0}$ $\nrightarrow$ $\lim_{t \rightarrow 3} (t+t^2) = 0$
 $\downarrow$ $\downarrow$ $\downarrow$
 integrate $t_0 = 3$ $\delta(t)$
 $\downarrow$ $\downarrow$
 $\delta = -2$ $t_0 = 3$ $\delta(t)$
 (Sifting Property)

b) $\int_{t=-2}^{t=4} \underbrace{(t+1)^2}_{x(t)} \underbrace{s(t-3)}_{t_0=3} dt$

$\delta(t-3)$

17a Sifting Property
↳ Integrate only at t_0
: $x(t_0=3) = x(3)$
: $3 + 3^2 = 12$

c) $\int_{t=0}^{t=3} \underbrace{e^{t-2}}_{x(t)} \delta(t-4) dt$

Scaling Property: $\int_{t=0}^{t=3} e^{t-2} \cdot \frac{1}{111} \delta(t-\frac{4}{2})$

$= \frac{1}{2} \int_{t=0}^{t=3} e^{t-2} \cdot \delta(t-2)$

$t_0 = 2$

Integrate includes $\therefore$ then $x(t_0)$ via Sifting Property

$\therefore x(t_0+2) = e^{2-2} = e^0 = 1$

$\therefore \int_{-1}^2 \underbrace{e^{t-2} \delta(t-4)}_{\text{ohhh!!}} = \frac{1}{2} \left(1 \right) \cdot \left(\frac{1}{2} \right)$

d) $\int_{t=-\infty}^{t=t_1} \delta(\tau) d\tau$

nona 2 di

$\delta(\tau)$

$\tau=0$

t_{L0}

t_{Z0}

Plot

1

0

τ

"Unit Step"

$t < 0: \int_{t=-\infty}^{t=t} \delta(\tau) d\tau = 0$ (im Range von Sifting property) $\therefore \text{Ans } \int_{-\infty}^{\infty} \delta(\tau) d\tau = \begin{cases} 0 & ; t < 0 \\ 1 & ; t > 0 \end{cases}$
 $t > 0: \int_{t=-\infty}^{t=t} \delta(\tau) d\tau = 1$ (von $\int_{-\infty}^{\infty} \delta(\tau) d\tau = 1$) $\Rightarrow$ Summe 1
 Mit $\frac{d}{dt} u(t) = \delta(t)$
 Lösung muss zu $t=0$ sein

Note $\frac{d}{dt} u(t) : \delta(t)$

↓

Unit Step

DISCRETE-TIME SIGNALS

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DISCRETE-TIME SIGNALS

- Signals that are defined only at discrete instants of time, i.e.,
 - number of children born on a specific day in a year
 - Number of new Covid-19 cases
- Signals occur when an analog signal is converted into a discrete-time signal by the process of sampling, i.e.,
 - Digital recording of audio signals
 - Telemetry system (data from several measurement sensors are transmitted over a single channel by time sharing)

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Note n consider Integer #

DISCRETE-TIME SIGNALS

$$E_x = \cos \frac{\pi}{6} t \quad T = \frac{2\pi}{\omega_0} = \frac{2\pi}{\frac{\pi}{6}} = 12 \text{ sec}$$

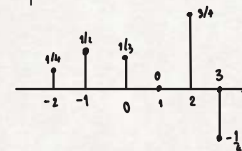
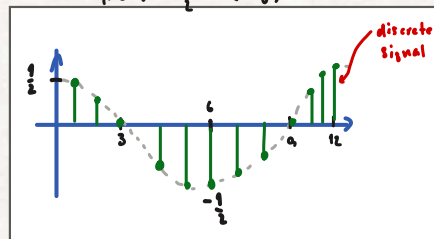

- We represent the discrete-time signal as a sequence of values $x[n]$ with n assuming only integer values. Note $n: 0, \pm 1, \pm 2$

↓
discrete signal

- Representing DT signals in functional form: $x[n] = \frac{1}{2} \cos(\pi n/6)$

- Representing DT in as the elements of sequence: $x[n] = \{\frac{1}{4}, \frac{1}{2}, \frac{1}{3}, 0, \frac{3}{4}, -\frac{1}{2}\}$

$$x[n] = \frac{1}{2} \cos \left(\frac{\pi n}{6} \right)$$



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ENERGY VS. POWER SIGNALS

- Energy of a DT signal $x(n)$, for some integer $N > 0$ Summation

$$E = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt$$

$$E = \lim_{N \rightarrow \infty} \left(\sum_{n=-N}^N |x[n]|^2 \right)$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

$x[n]$ IS AN ENERGY SIGNAL IF E IS FINITE, AND IT IS A POWER SIGNAL IF E IS NOT FINITE, BUT P IS FINITE

$$E_x x[n] = \cos \left(\frac{\pi}{4} n \right)$$

- The signal $x[n]$ is periodic if, for some integer $N > 0$, $x[n+N] = x[n]$ for all n

$$x[n+N] = x[n] \text{ for all } n$$

- The smallest value of N that satisfies this relation is the fundamental period of the signal

- If there is no integer N that satisfies the equation, $x[n]$ is an aperiodic signal

Note Discrete Signal $x[n] = A \sin(2\pi f_0 n + \phi)$

$$x[n+N] = A \sin(2\pi f_0 (n+N) + \phi) = A \sin(2\pi f_0 n + \phi + 2\pi f_0 N)$$

$$A \sin(2\pi f_0 n + \phi) = A \left[\sin(2\pi f_0 n + \phi) \cos(2\pi f_0 N) + \cos(2\pi f_0 n + \phi) \sin(2\pi f_0 N) \right]$$

isolated LHS: RHS also

$$\cos(2\pi f_0 N) = 1 \text{ \& \; } \sin(2\pi f_0 N) = 0 \Rightarrow f_0 N = \text{min Integer.}$$

$$E_x x[n] = \cos \left(\frac{\pi}{4} n \right)$$

$$\omega_0 = \frac{\pi}{4} \Rightarrow f_0 = \frac{\omega_0}{2\pi} = \frac{\pi}{4} \cdot \frac{1}{2\pi} = \frac{1}{8}$$

$$\frac{1}{8} N = \text{min Integer} \rightarrow N = 8, \text{ min } 2\pi = 1$$

Lo x do Sampling at 18 on 16000000

$$E_x x[n] = x_1[n] + x_2[n] \rightarrow T_1, T_2$$

$$x[n+N] = x_1[n+N] + x_2[n+N]$$

$$N = p N_1 = q N_2 \rightarrow \text{should be even Continuous}$$

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TRANSFORMATIONS OF INDEPENDENT VARIABLE

- For integer value of k , sequence $x[n-k]$ represents $x[n]$
shifted

- Shifted by k samples

- Shift to the right if $k > 0$ Ex $x[n-3]$

- Shift to the left if $k < 0$ Ex $x[n-(-3)]$

Note $x[n] \rightarrow x[an+b]$

if $|a| > 1 \Rightarrow$ compressed (unusual)

if $0 < |a| < 1 \Rightarrow$ expanded

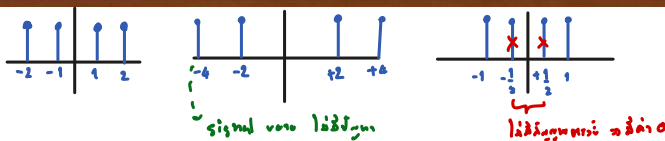
- $x[-n]$ corresponds to reflecting $x[n]$ at around origin $n = 0$

Continuous Sig.

- Scaling is similar to CT signal, but must be interpreted with care in discrete-time case, since the signals are defined only for integer values of the time variable.

!! Γ: κλίμακα αποκλίμακα αποκλίμακα αποκλίμακα $\therefore$ compressed αποκλίμακα integer !!

$$x[n] \xrightarrow{\text{expand}} x\left[\frac{n}{2}\right] \xrightarrow{\text{compress}} x[2n]$$



TRANSFORMATIONS OF INDEPENDENT VARIABLE

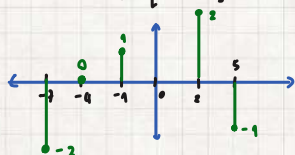
EXERCISE



Consider the waveform shown in the figure on the right, and let

$$y[n] = x\left[-\frac{n}{3} + \frac{2}{3}\right]$$

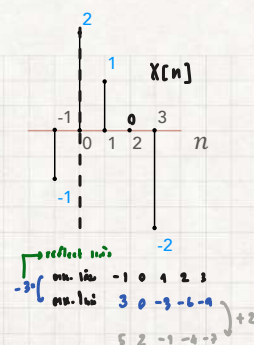
Scale down $y[n] = -\frac{1}{3}x[n-2]$ $\Rightarrow$ scale 3 shift right 2



Sequence: $\{-1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1\}$

(111110000)

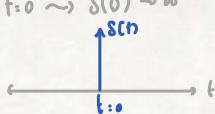
$\Rightarrow$ scale 3 shift right 2 scale 3 shift right 2 scale 3 shift right 2



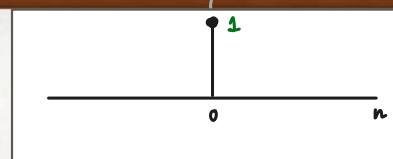
Note: scale 3 shift right 2 scale 3 shift right 2 scale 3 shift right 2

Continuous သာ $\delta(t) \rightarrow \infty$
Discrete သာ $\delta[n] = 1$

Note: $\delta(t) @ t=0 \rightarrow \delta(0) \rightarrow \infty$
Unit Impulse Function
[Continuous]



ELEMENTARY SIGNALS



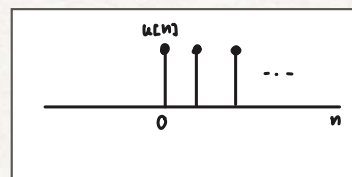
- Unit Impulse Function in discrete time is defined as

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

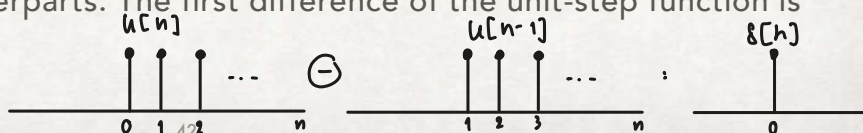
- $\delta[n]$ is somewhat similar to the continuous-time impulse function $\delta(t)$ we note that the magnitude of the discrete impulse is always finite.

- Unit-step sequence is defined as

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



- The discrete-time delta and step functions have properties somewhat similar to their continuous-time counterparts. The first difference of the unit-step function is $u[n] - u[n-1] = \delta[n]$



(Discrete Delta)

ELEMENTARY SIGNALS $\Rightarrow$ သာသနာ $\approx$ Continuous

Other properties

Delta Discrete $\longleftrightarrow$ Continuous

$$\sum_{k=-\infty}^n \delta[k] = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases} = u[n]$$

Unit Impulse? for Discrete

$$\sum_{k=0}^{\infty} \delta[n-k] = u[n]$$

$$\int_{-\infty}^t \delta(\tau) d\tau = u(t)$$

$$\int_0^{\infty} \delta(t-\tau) d\tau = u(t)$$

Similar properties to the continuous-time system:

$$x[n]\delta[n-k] = x[k]\delta[n-k] \longleftrightarrow x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$$

သော $t=t_0$

General form:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \longleftrightarrow x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau) d\tau$$

represent signal as summation of impulses

$$x[n] = a\delta[n-n_1] + b\delta[n-n_2] + c\delta[n-n_3]$$

n_1, n_2, n_3

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Summary ; Continuous Signals VS Discrete Signals

Periodic ?

VS